Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist, *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, identifying $1.5M+ in expansion opportunities for clients.
- Built a revenue driver tree (traffic × capture rate × average ticket) and margin models that turned client profit goals into break-even targets.
- Estimated the incremental revenue of new sites with observational causal inference (Double ML, causal forests) on a 12-year county panel.
- Checked the causal estimates with difference-in-differences and out-of-sample back-tests against operating sites.
- Automated Python pipelines for Monte Carlo risk and low/base/high scenarios (150+ unit tests, data-quality checks), cutting report turnaround 91%.

Senior Data Scientist, *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Developed uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, cutting ad spend by $10K per campaign.
- Built an in-house experimentation and reporting platform that data science and strategy used to validate campaign lift, automate metric monitoring, and feed Power BI dashboards.
- Identified insurance claims and support-ticket complaints as the strongest leading signals of member churn and added them to the churn model.
- Re-architected the churn pipeline on Spark and Databricks (partitioning, caching, removing repeated joins), cutting runtime by 48% and memory by 23%.
- Built a certified gross lead rate (GLR) definition for a new client whose data lacked key lead markers, working with data engineering to date leads by last response, credit them to campaigns, and flag broker-influenced leads.

Founding Machine Learning Engineer, *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, built the search ranking and shipping price models, judged on NDCG@10 and MAE, adding $420K in annualized profit.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) that cut erroneous and outlier shipping prices by 13%, reducing cost leakage while maintaining an 18% contribution margin.
- Built automated drift monitoring for live inference pipelines and mentored the software engineering team on production ML practices.
- Cut search complaints and negative feedback by 12% and raised search engagement by 16% with custom Elasticsearch analyzers, field boosting, and relevance tuning.

Data Scientist, *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Improved the XGBoost GLR model's AUC-ROC by 3% relative to the prior model, confirmed it with an A/B test, and used SHAP to identify the main drivers (income, age, household composition).
- Used uplift models to estimate creative-driven lift in GLR, and validated GLR out of sample as a 60-day leading indicator of member rate ($R^2 = 0.28$).
- Raised GLR by 18% (relative) through model-driven targeting and creative changes.
- Improved geospatial targeting precision by 20% with GeoPandas GIS analysis, shifting campaign resources to high-opportunity regions.
- Led monthly monitoring of up to 10 production client models (accuracy, feature drift, forecast vs. actual) and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes with tenure-dependent churn (10%, 5%, then 2% monthly) to show the LTV = ARPU / churn identity holds only under constant churn.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), matching the simulated mean.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members than a prior-year-cost baseline.

EDUCATION

Ph.D. in Mathematics, *University of Pittsburgh* 2017 - 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics, *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Causal Inference & Experimentation: A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Observational Causal Methods, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs), Robustness Checks

Modeling & Statistics: Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, SHAP, Drift Monitoring

Data & Tools: SQL, Python (pandas, statsmodels, scikit-learn, econml), R (survey analysis), C#, Snowflake, Databricks, Spark, dbt, Curated Tables & Data Marts, AWS, Azure, Power BI, Git

Collaboration: Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring

Metrics & Planning: KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Consumer Product Analytics (Acquisition, Activation, Engagement, Retention, Monetization), CAC, LTV / ARPU, Contribution Margin, Scenario Analysis, Target Setting, Budget Allocation, Forecasting